STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 6: System of Particles and Rotational Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Mass-weighted mean position.

Suggested allocation by section: 1-mark questions earn one mark for the correct fact, formula or unit; 2-mark questions normally award one mark for the principle/formula and one for the explanation or result; 3-mark questions award credit for the relation, working/reasoning and result with unit; 4-mark questions award credit for the principle, equation, working and final answer/interpretation. Accept equivalent correct methods.

Q2 1 mark

rF sintheta.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q3 1 mark

kg m^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q4 1 mark

Iomega^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q5 1 mark

Net external torque is zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q6 1 mark

v=romega.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q7 1 mark

sumF=0 and sum=0.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q8 1 mark

I=I_cm+Md^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q9 2 marks

3 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q10 2 marks

2 N m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q11 2 marks

24 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q12 2 marks

20 rad/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q13 2 marks

3 rad/s^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q14 2 marks

Constant.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q15 3 marks

3 m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q16 3 marks

5 N m.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q17 3 marks

I=.5 kg m^2; K=25 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q18 3 marks

2 rad/s^2; 24 rad.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q19 4 marks

3 m; 2 m/s^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q20 4 marks

omega=7.5 rad/s; K increases 18 to 45 J because skater does work.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.